

Pooja Sethi

Research Engineer & Technical Lead

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Summary

Research Engineer & Technical Lead specializing in LLM post-training, data curation, and evaluation at scale. Proven track record developing high-reliability benchmarks, evaluation infrastructure, and multilingual models that drive downstream product impact at Meta, serving billions of users daily. Deep experience in NLP, machine translation, and research-to-production pipelines.

Experience

- Oct 2022–Present **Senior Research Engineer & Technical Lead, Machine Translation, Meta Content Understanding**, Seattle, WA
- Technical lead for 10+ person team developing Meta's translation strategy, replacing legacy neural machine translation models (NMT) with LLM-based systems serving O(billion) translations daily across Instagram and Facebook.
 - Spearheaded research-to-production efforts for FAIR's Seamless speech-to-speech translation system. Led post-training and speech-to-text evaluation across 16 languages and shipped voice dubbing feature in Instagram Reels (announced by Mark Zuckerberg at Meta Connect 2024).
 - Shipped 8b translation model to 100% of Instagram Reels Captions translation traffic based on results showing quality gains while meeting latency/cost constraints, resulting in statistically significant improvements to top-line engagement metrics like dwell time and reshares.
 - Developed low-resource translation models competitive with Gemini on Facebook Posts on priority low resource languages (Indic and African), via continuous pre-training and post-training.
 - Led data curation and post-training of LLMs for production text translation. Curated high-quality multilingual training data and designed evaluation methodology comparing distilled models against teacher models on quality versus cost.
 - Built comprehensive evaluations for multilingual LLMs spanning 100 languages. Developed automated benchmark suite combining translation quality metrics (BLEU, MetricX, chrF), human evaluation pipelines, and A/B testing to drive model selection.
- Oct 2021–July 2022 **Senior Software Engineer 2, Machine Learning, Impira (Acquired by Figma)**, San Francisco, CA
- First NLP engineer at Series B startup (30 people), reporting to CTO.
 - Developed and shipped Vision Transformer-based models (LayoutLM) for unsupervised document clustering and classification (using text + vision features) to enable information extraction.
 - Created benchmark datasets and evaluation metrics for document similarity and clustering quality.
 - Research demo and evaluation methodology documented at github.com/poojasethi/visual-doc-clustering.
- Oct 2017–Oct 2021 **Senior Software Engineer, Machine Learning, Facebook Reality Labs**, Redmond, WA
- Early engineer on Natural Language Understanding team for Facebook Assistant (Portal, Oculus, Ray-Ban Meta devices).
 - Built AutoNLU, an active learning and evaluation platform for semantic parsing. Developed error detection algorithms identifying model failure modes and root-cause analysis tooling.
 - Designed and implemented evaluation pipelines for intent classification and slot filling models, including adversarial test sets and cross-domain generalization benchmarks.
 - Conducted 100+ technical interviews. Bootstrapped and grew internal mentorship program to 70 participants.

Research Experience

- Sept 2016–Mar 2017 **Research Assistant, UW Natural Language Processing (NLP) Group, Seattle, WA**
- Proposed new NLP task: conversational question generation (CQG). Work received Facebook ParIAI research award (\$19,800).
 - Developed RNN-based question generation models with evaluation framework including automated metrics (BLEU, ROUGE) and human evaluation protocols.
 - Advised by Professor Yejin Choi.
- Apr 2014–June 2017 **Research Assistant, UW ICT for Development (ICTD) Lab, Seattle, WA**
- Developed Respeak, an Android crowdsourcing application for low-income blind users in India.
 - Created evaluation benchmarks demonstrating 6X improvement in Hindi and 3X improvement in Indian-accented English ASR accuracy.
 - Designed user study methodology and quality metrics for voice-based task completion. Published evaluation results at CHI (Best Paper Honorable Mention).
 - Advised by Professor Richard Anderson and Aditya Vashistha.
- Mar 2015–June 2015 **Visiting Undergraduate Researcher, Allen Institute for AI (AI2), Seattle, WA**
- Explored adversarial perturbations of science questions to systematically increase or decrease difficulty for Q&A models.
 - Developed evaluation framework for measuring model robustness and identifying failure modes.
 - Advised by Been Kim (now at Google Brain).

Education

- 2021–2025 **M.S. in Computer Science, AI Concentration, Stanford University, GPA: 3.94/4.0**
Honors Cooperative Program
Select Coursework: Deep Reinforcement Learning, Trustworthy Large Language Models, Convex Optimization, Deep Multi-task and Meta-learning, Decision Making Under Uncertainty, NLP, Computer Vision, ML Under Distribution Shifts.
- 2013–2017 **B.S. in Computer Engineering, University of Washington, Seattle, GPA: 3.85/4.0**
Magna Cum Laude with Department Honors
Awards: Best Undergraduate Honors Thesis Award; Outstanding Computer Engineering Senior Award (top two graduating seniors).

Technical Skills & Expertise

- ML/AI LLM post-training & evaluation, RLHF, multilingual NLP, machine translation, speech processing, model distillation, active learning, transformer architectures (Llama, LayoutLM, encoder-decoder models)
- Data Curation Large-scale dataset creation and curation, multilingual data collection, organic and synthetic data generation, data quality assessment, Hive and ETL pipelines
- Evaluation Benchmark design and curation, automated evaluation pipelines, human evaluation protocols, A/B testing, quality metrics (BLEU, MetricX, chrF), adversarial testing
- Engineering PyTorch, Python, HuggingFace trainers, distributed training, research-to-production workflows

Selected Publications

- Preprint **AutoNLU: Detecting, root-causing, and fixing NLU model errors.** Pooja Sethi, Denis Savenkov, et al.
- CHI 2018 **BSpeak: An Accessible Voice-based Crowdsourcing Marketplace for Low-Income Blind People.** Aditya Vashistha, Pooja Sethi, and Richard Anderson.

CHI 2017 **Respeak: A Voice-based, Crowd-powered Speech Transcription System.** Aditya Vashistha, Pooja Sethi, and Richard Anderson. (*Best Paper Honorable Mention - top 5%*)

Selected Awards & Honors

- 2017 Facebook ParIAI Research Grant (\$19,800 for conversational question generation research)
- 2017 Outstanding Computer Engineering Senior Award (top graduating senior)
- 2017 Best Undergraduate Honors Thesis Award
- 2013 UW CSE Freshman Direct Admit (<5% acceptance rate)
- 2013 Admitted to MIT class of 2017, Early Action (declined)